

Virological Determinants of Human Cytomegalovirus Transmission

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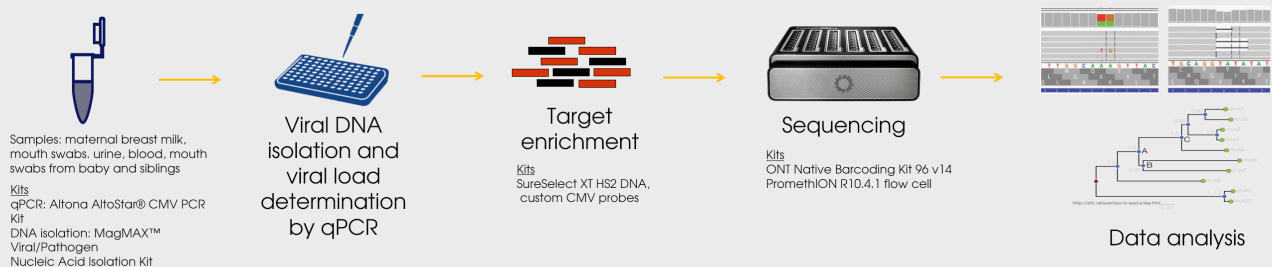
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Background: Human cytomegalovirus (HCMV), a ~236 kbp DNA virus in the Herpesviridae family, is highly prevalent (50–100% seroprevalence) and infection is typically asymptomatic but poses severe risks to immunocompromised individuals and developing fetuses. Congenital HCMV (cCMV), the leading cause of non-genetic childhood hearing loss, affects 0.2–2% of live births and imposes a significant economic burden (~\$2 billion/year in the U.S.). Despite the urgent need, no effective vaccine exists, largely due to the extensive genetic diversity of HCMV. To enhance our understanding of CMV genetic diversity, robust methods are required for comprehensive genomic analysis of its 236 kbp genome from clinical samples.

Hypothesis: We hypothesize that CMV transmission selects for specific viral genotypic features that confer a selective advantage. Identifying these features will provide potential targets for a CMV vaccine.

Objective: Our goal is to develop experimental and analytical methods for the genomic profiling of complete CMV genomes, enabling precise characterization of CMV strains present in a sample.

Methods



Results

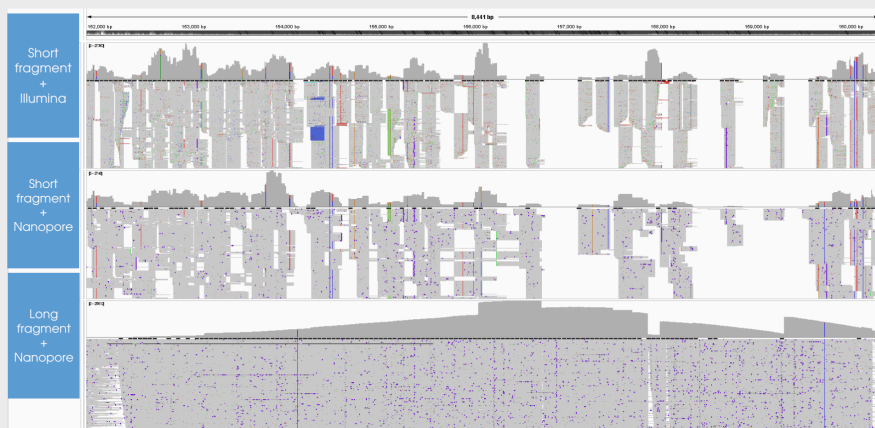


Figure 1. Gaps in the alignments resolved by the modified long fragment protocol. Coverage depths showing low-depth regions or gaps resolved by the modified long fragment protocol. Top panel shows the alignment of the short fragment (SF) +Illumina data, middle panel is SF+Nanopore, bottom panel is Long fragment+Nanopore. Alignments were visualized using IGV.

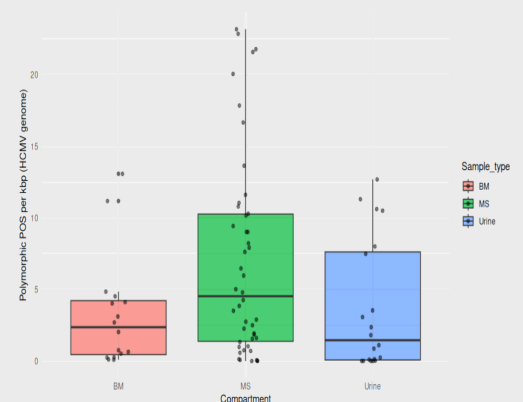


Figure 2. Polymorphic positions per kbp per sample type. Data indicates that only ~4% of the HCMV genome is polymorphic. MS, mouth swabs. BM, breast milk.